Restricting Range Restricts Conclusions: Studying the Best Can Mislead

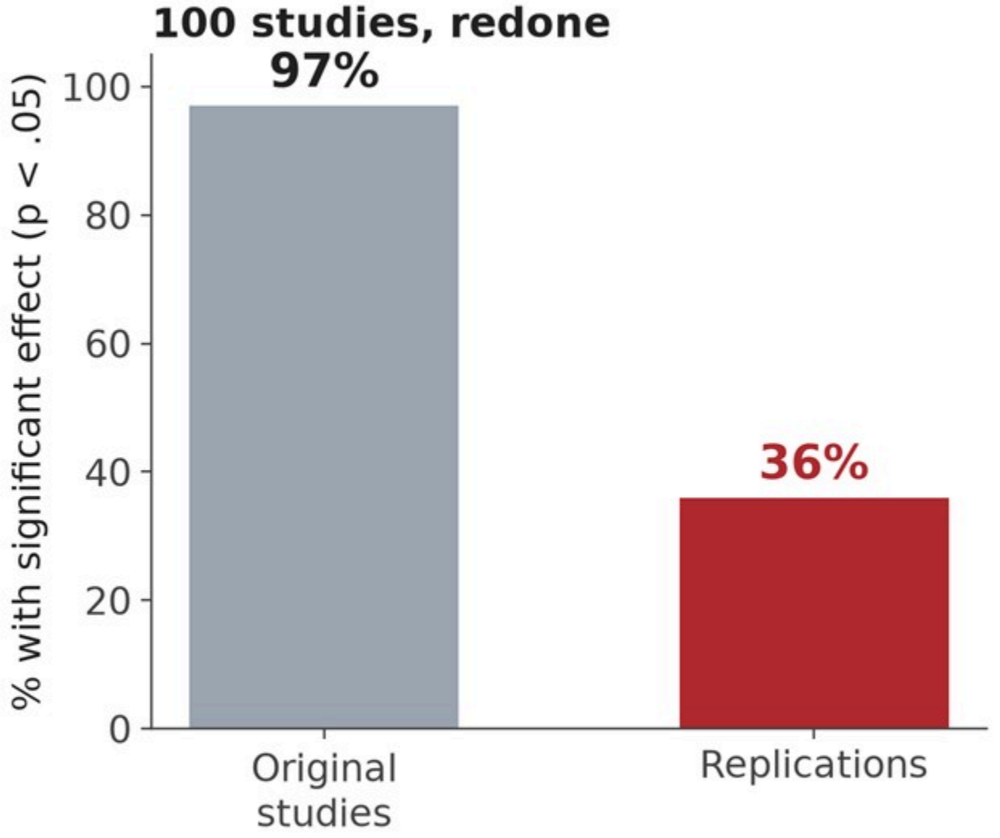
Selection Bias in Expertise Research

Dr Nemanja Vaci



Psychology's stress test

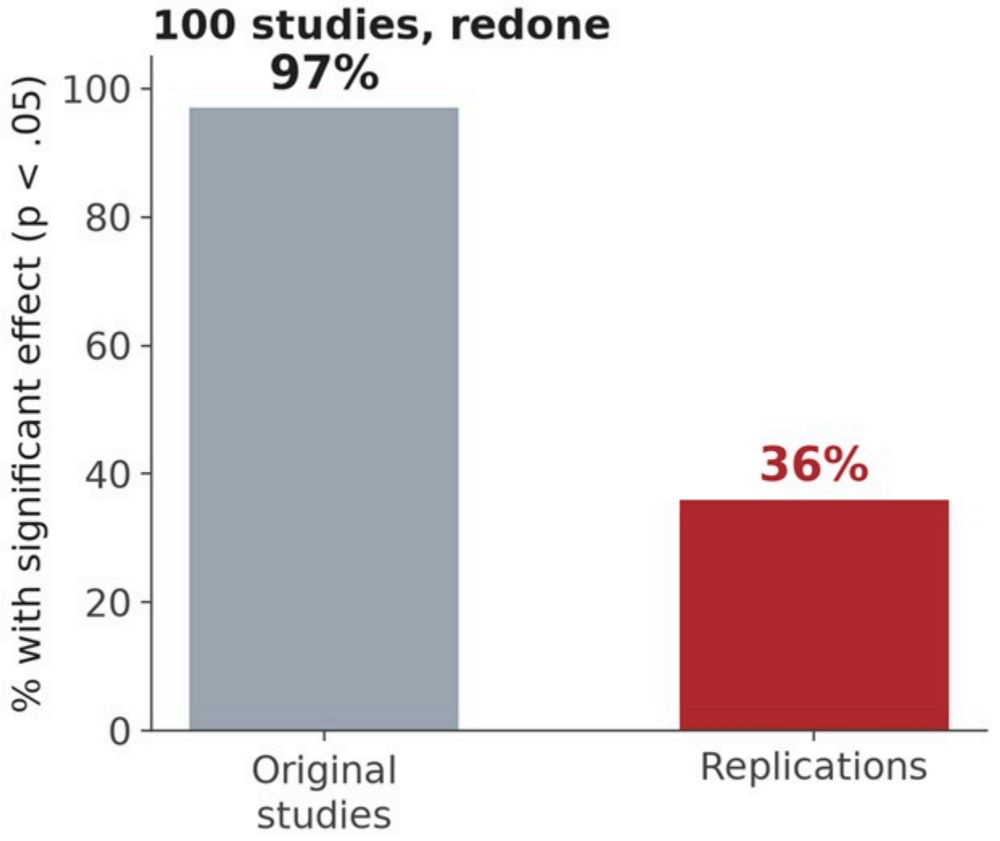
Open Science Collaboration (2015, Science): 100 published studies, independently redone

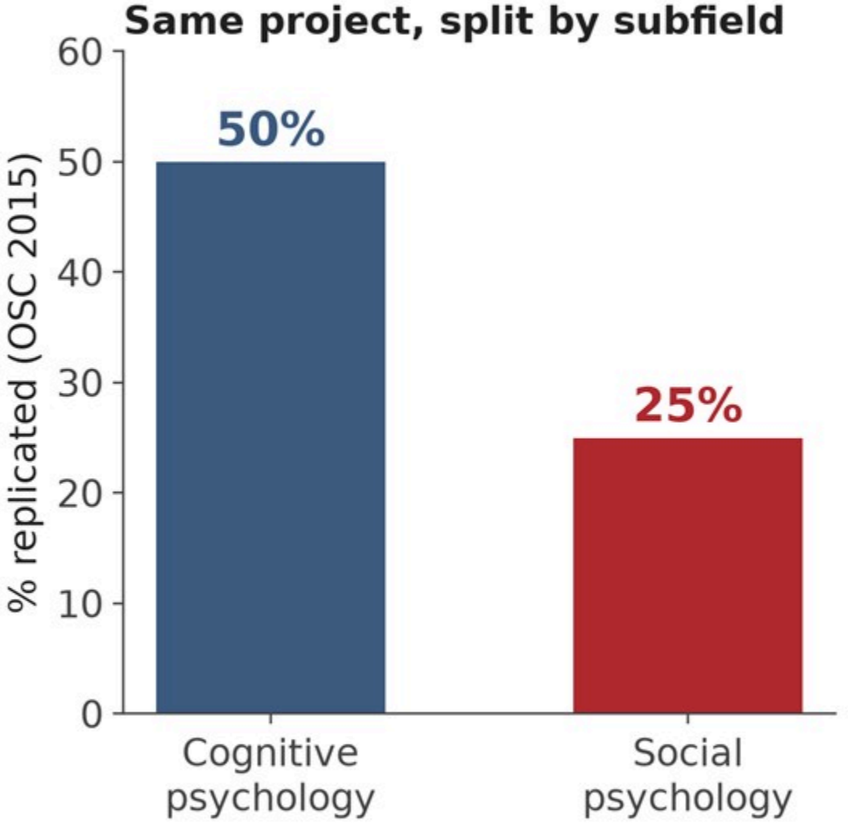


36% of replications significant with effects shrinking to half their size.

Psychology's stress test

Open Science Collaboration (2015, Science): 100 published studies, independently redone





36% of replications significant with effects shrinking to half their size.

controlled, within-task experimental effects survive;
correlational and contextual effects are fragile.

Expertise research has both traditions in one field

Expertise as PREDICTOR

What is expertise? (experimental)

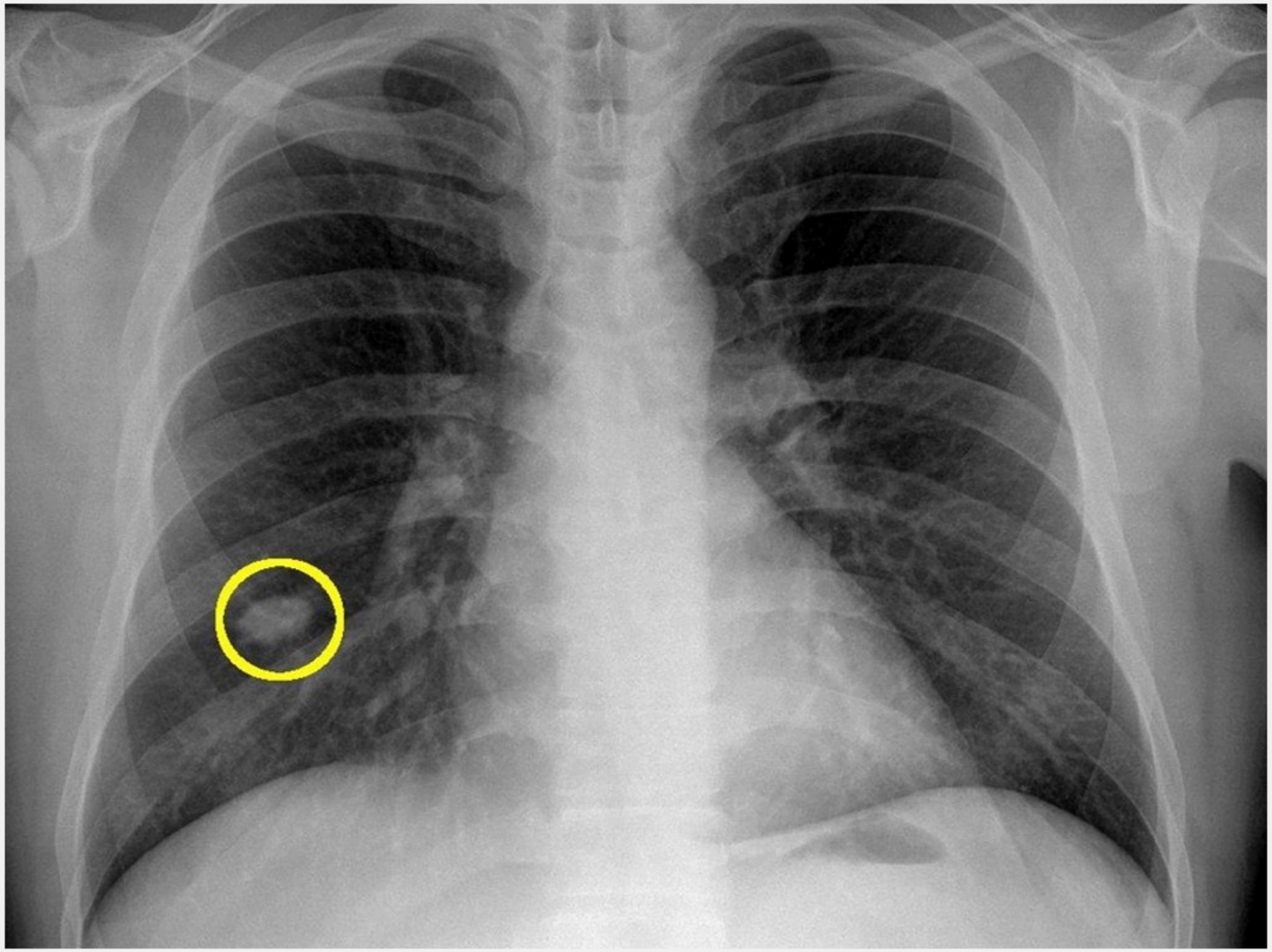
Design: expert–novice comparisons under controlled conditions

Logic: expertise is the independent variable; mechanisms are the outcome

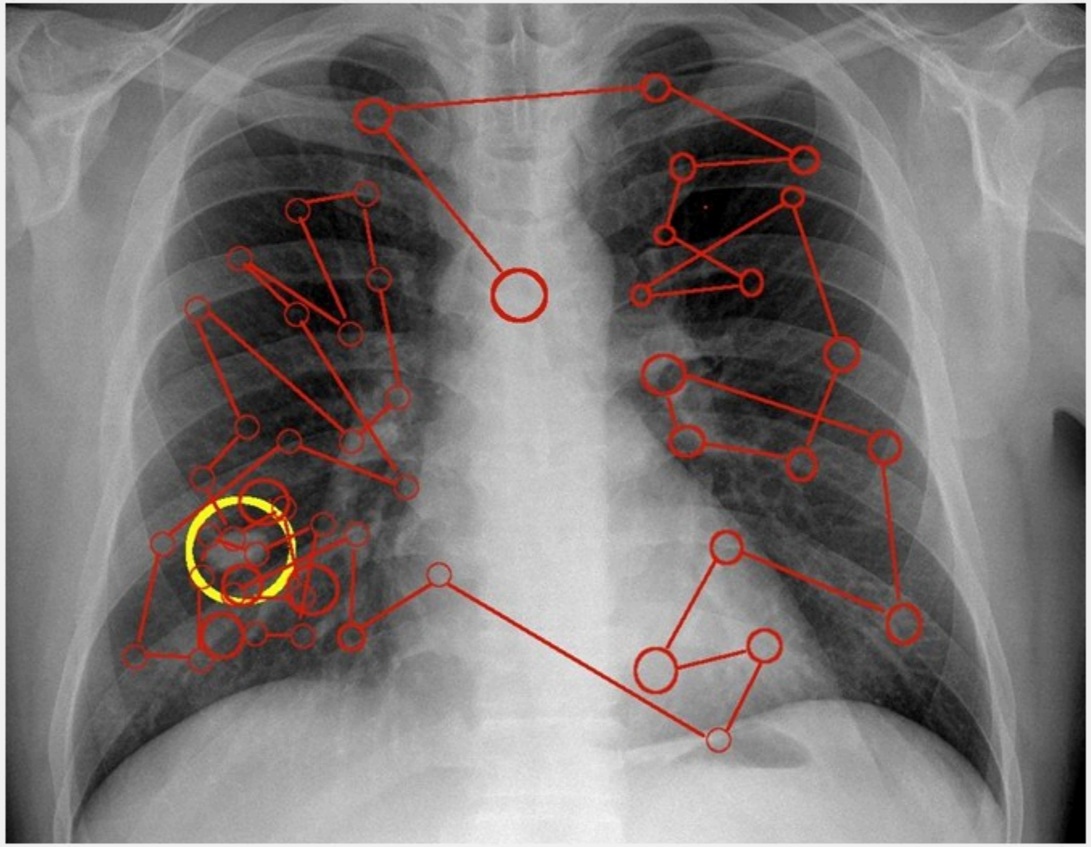
Findings: Selective search (de Groot, 1946), Chunking & templates (Chase & Simon, 1973; Gobet & Simon, 1996), pattern recognition (Bilalić, 2017),

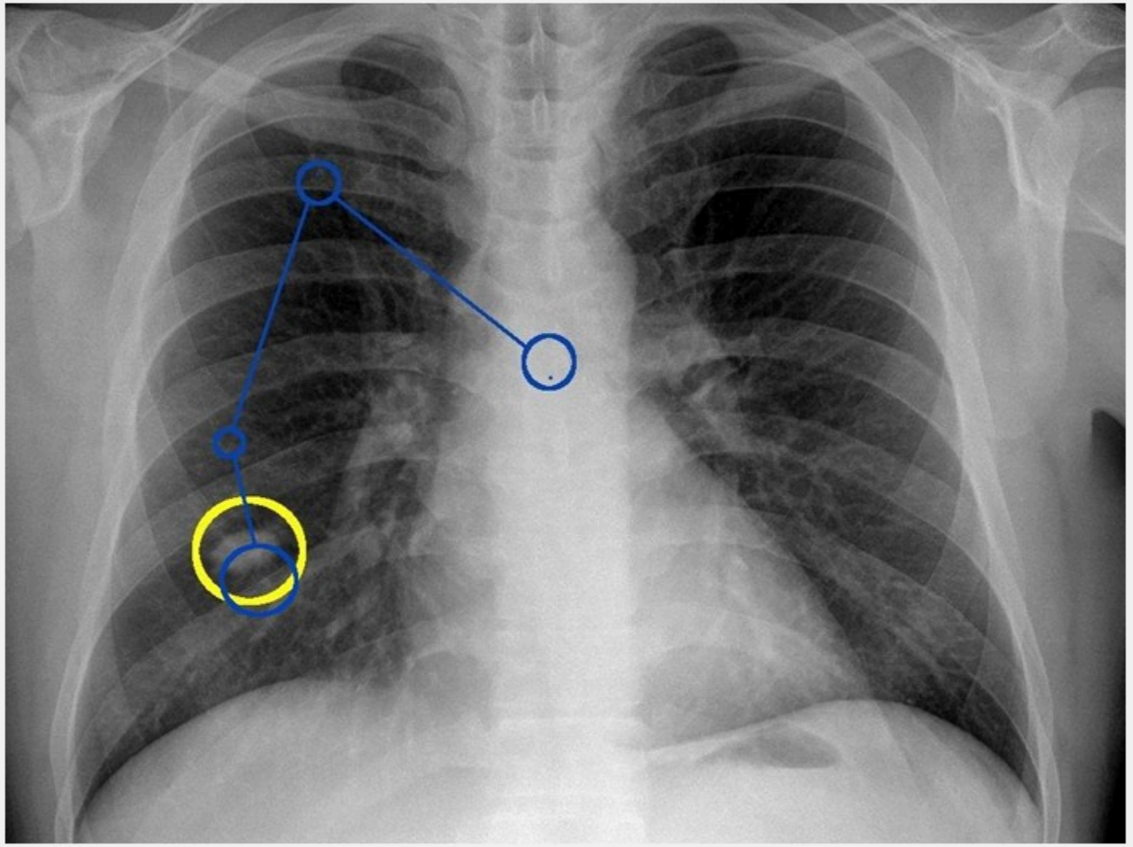
Theoretical predictions: precise, see CHREST and similar cognitive architectures

Is there a suspicious tissue in a scan?



Who is the expert?





Expertise research has both traditions in one field

Expertise as PREDICTOR

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Expertise as OUTCOME

How does expertise develop? (correlational)

Design: retrospective, prospective ,archival & survey studies of successful performers; experts vs everyone else

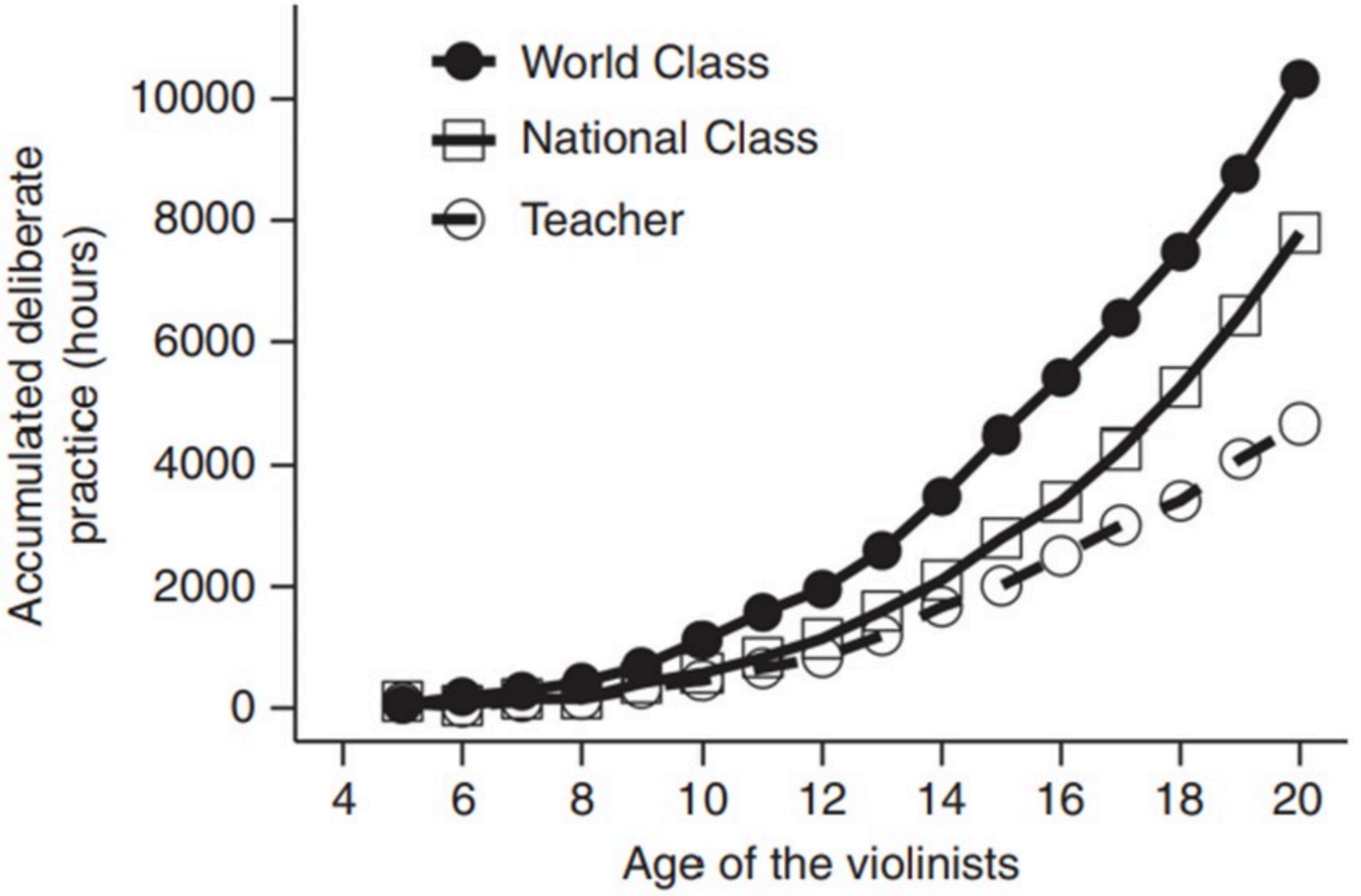
Logic: expertise is the dependent variable; early predictors are the cause

Findings: power-law of practice (Snoddy, 1928; Crossman, 1959); deliberate practice (Ericsson et al., 1993); early specialization (Güllich et al., 2022, 2025), cognitive abilities (Macnamara et al., 2014, 2016; Vaci, 2019)

Theoretical predictions: power-law and deliberate practice, everything else weak

Deliberate practice

1. Specific goals: improving well-defined micro behaviour
2. Immediate feedback: continuous and accurate feedback from coaches
3. Evaluation of the outcomes
4. Opportunities for repeated practice



The development of expertise

1993

Deliberate practice

Ericsson, Krampe & Tesch-Römer: structured effortful training as THE main factor

2014–16

The meta-analytic check

Macnamara et al.: practice explains far less — and less still in more selective samples

2017–19

Longitudinal modelling

Vaci & Bilalić (2017); Vaci et al. (2019, PNAS):
intelligence × practice jointly shape trajectories across the lifespan

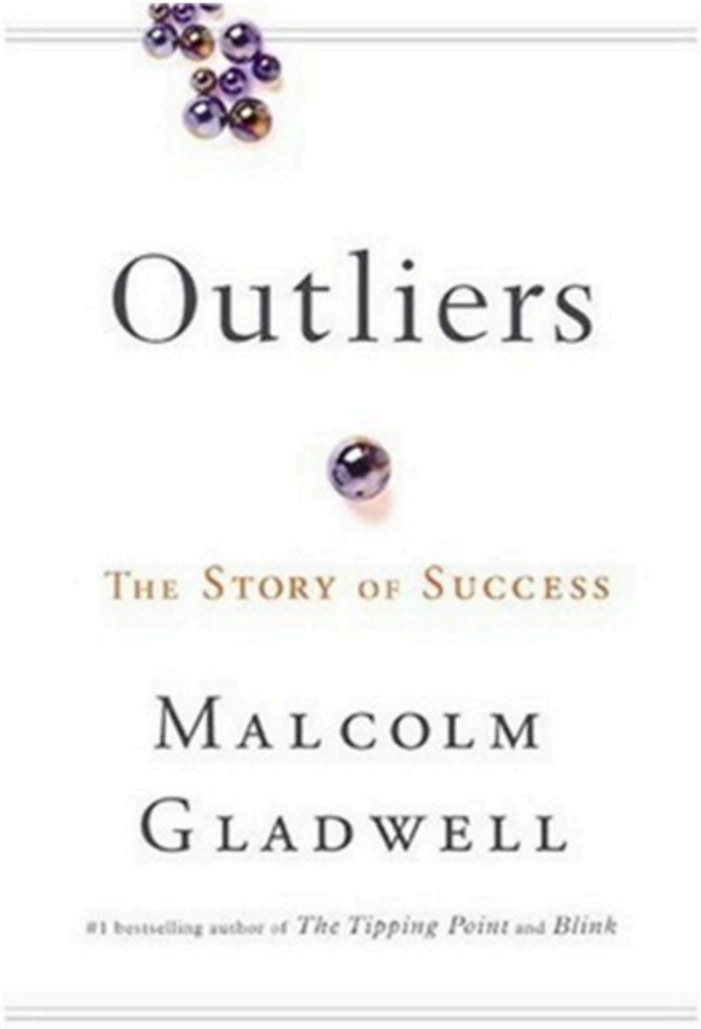
2022–25

The reversal claims

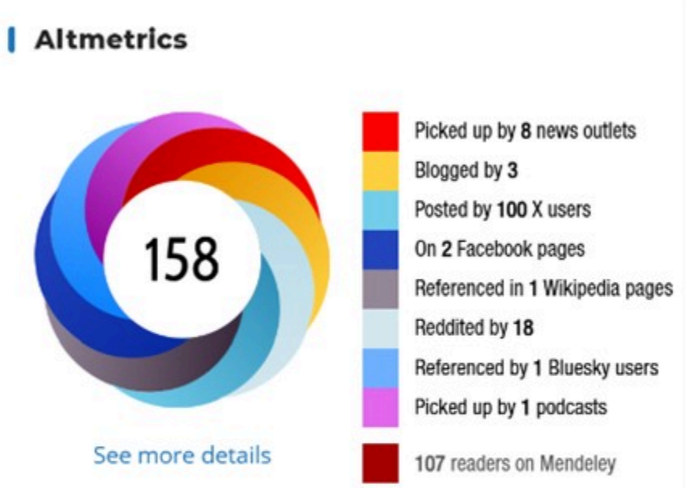
Güllich, Barth, Macnamara & Hambrick (Science, 2025):
among world-class adults, early specialization and early progress look harmful

Predictors that work in broad samples weaken or reverse among elites.

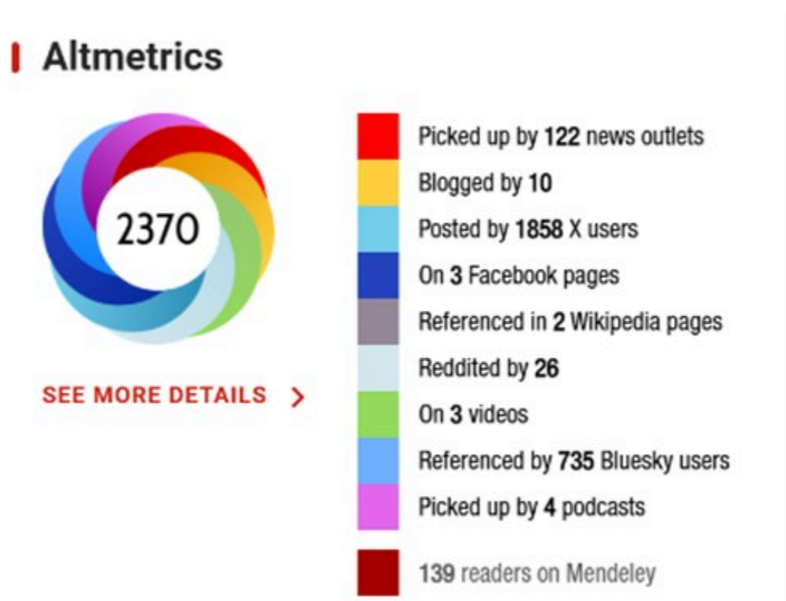
Stickiness of the results and claims



10 000 hours rule: Ericsson et al



Vaci et al: Numerical abilities x practice



Güllich et al: Late vs early specialization in expertise

Statistical artefacts induced by selection bias

Bilalić, Vaci, Hönekopp, & Pollet (in press). Studying the Best Can Mislead: Selection Bias in Expertise Research. Perspectives on Psychological Science.

Statistical artefacts induced by selection bias

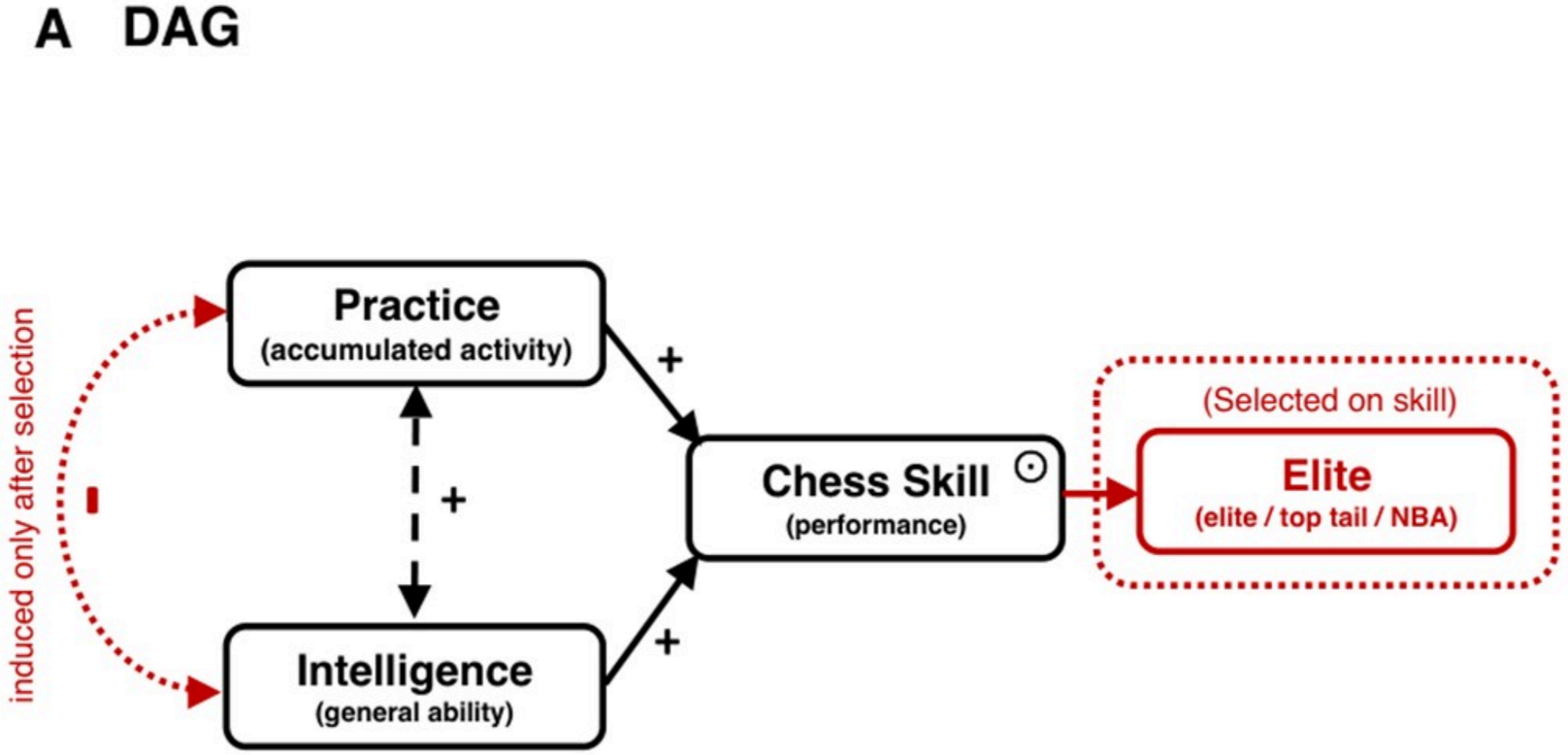
Elite (expert) samples are participants who have survived participation, opportunity, retention, and performance thresholds

Collider bias, range restriction, survivorship, and sampling variability

1. **Previous publications:** Directed Acyclic Graph (DAG) to illustrate selection mechanism
2. **Simulations:** can selection alone can reproduce key patterns in literature
3. **Empirical test**: validate simulations
4. **What to do about it:** Theoretical and practical considerations

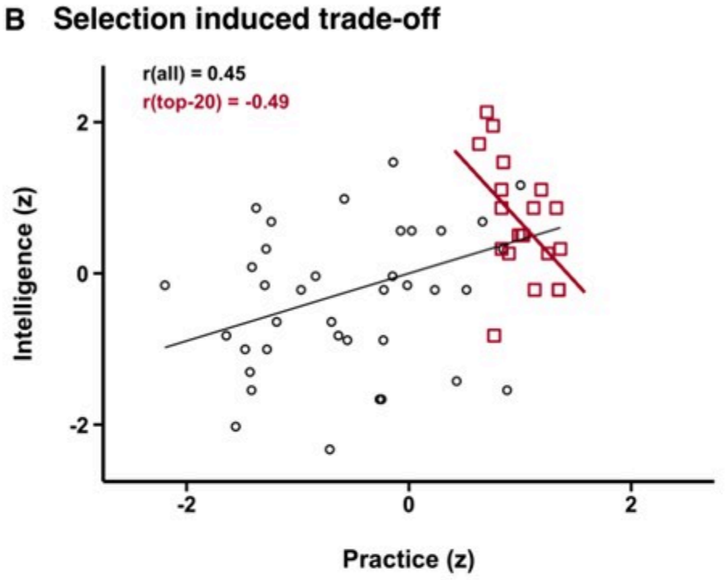
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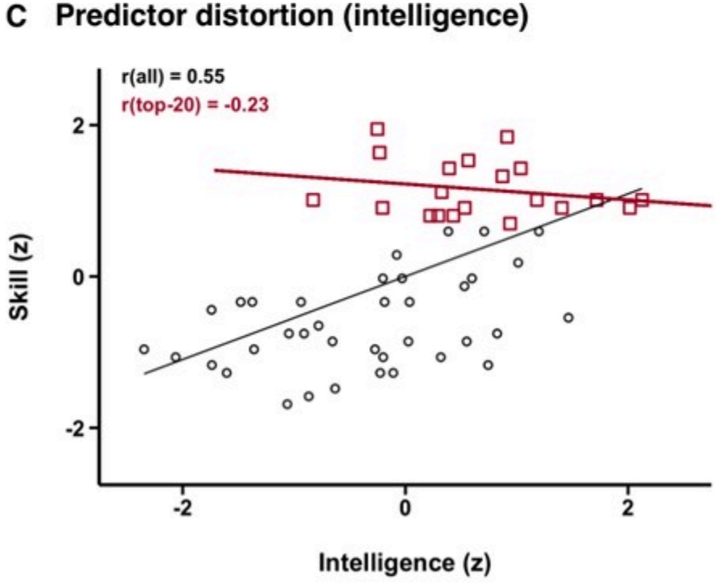
Previous work: Chess DAG

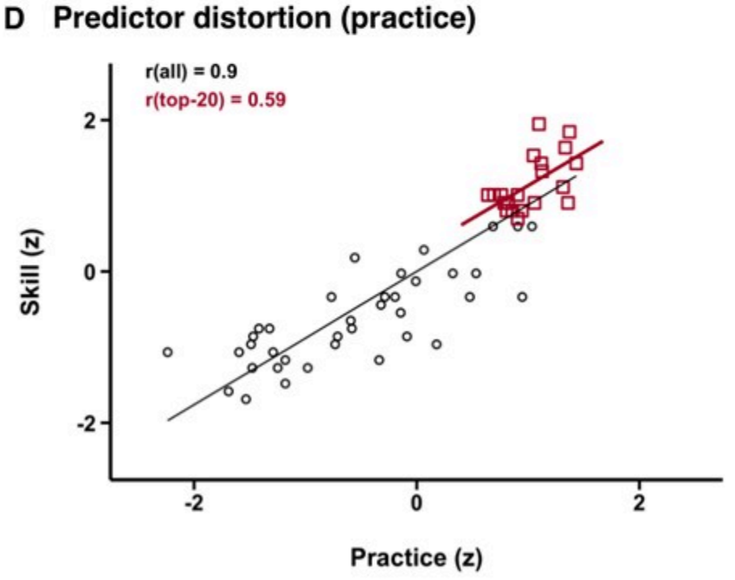


Bilalić, McLeod & Gobet (2007) — intelligence, practice, skill

Chess DAG: Conditioning on success







Full sample:

intelligence–practice r = .45

intelligence–skill r = .55

practice–skill r = .90

Top 20 by skill:

intelligence–practice r = −.49

intelligence–skill r = −.23

practice–skill r = .59

Selection alone created a trade-off, flipped one predictor, and attenuated the other.

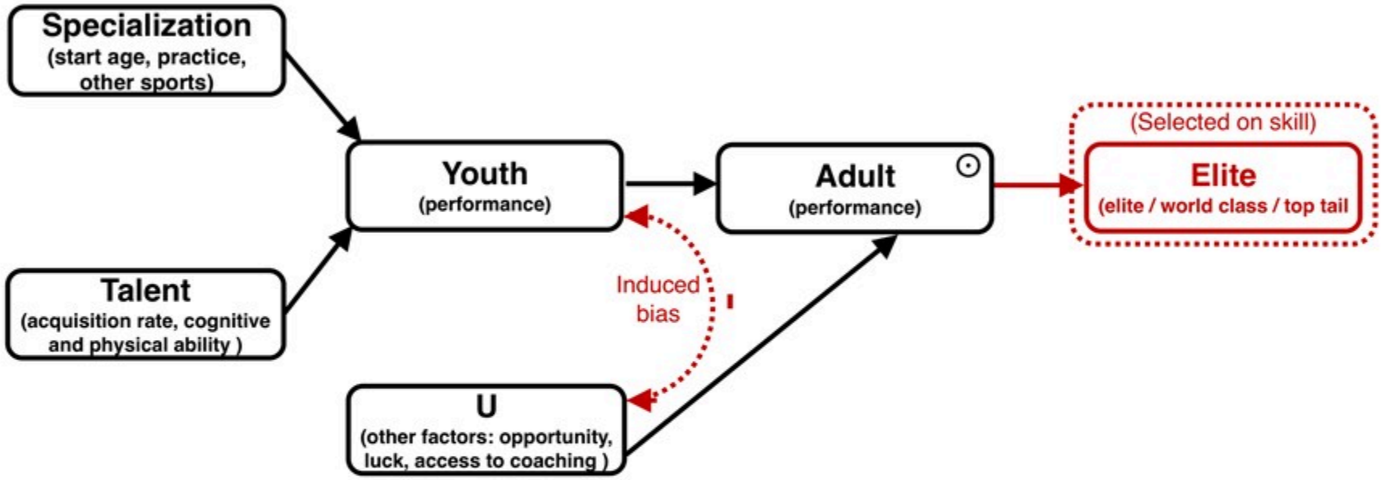
Simulation: A two-gate model of expertise development

Specialization + talent → youth performance → survival gate → adult performance → elite cutoff

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Specialization + talent → youth performance → survival gate → adult performance → elite cutoff

A Causal structure



Two factors: specialization and talent influencing youth performance

Adulthood works only through youth performance

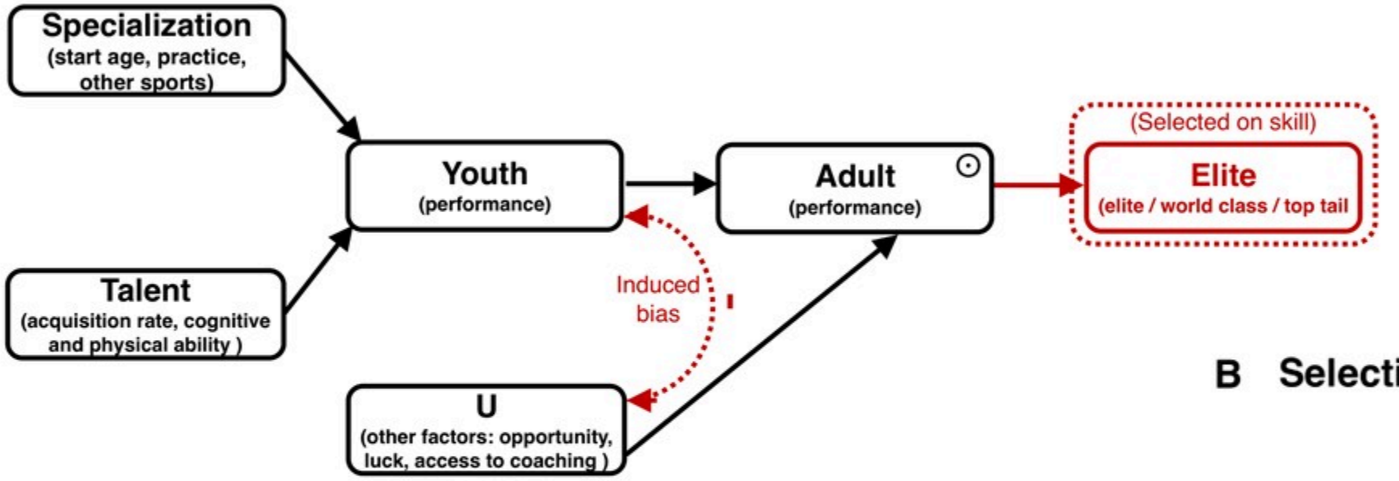
Later-emerging factors **U** (coaching, opportunity, maturation, luck) shape adult performance

Within elites: Youth performance and U must trade off

Simulation: A two-gate model of expertise development

Specialization + talent → youth performance → survival gate → adult performance → elite cutoff

A Causal structure



Gate 1 — survival: better youth performers stay visible (100k → 33k)

Gate 2 — elite cutoff: top-K by adult performance (33k → 100; 0.3%)

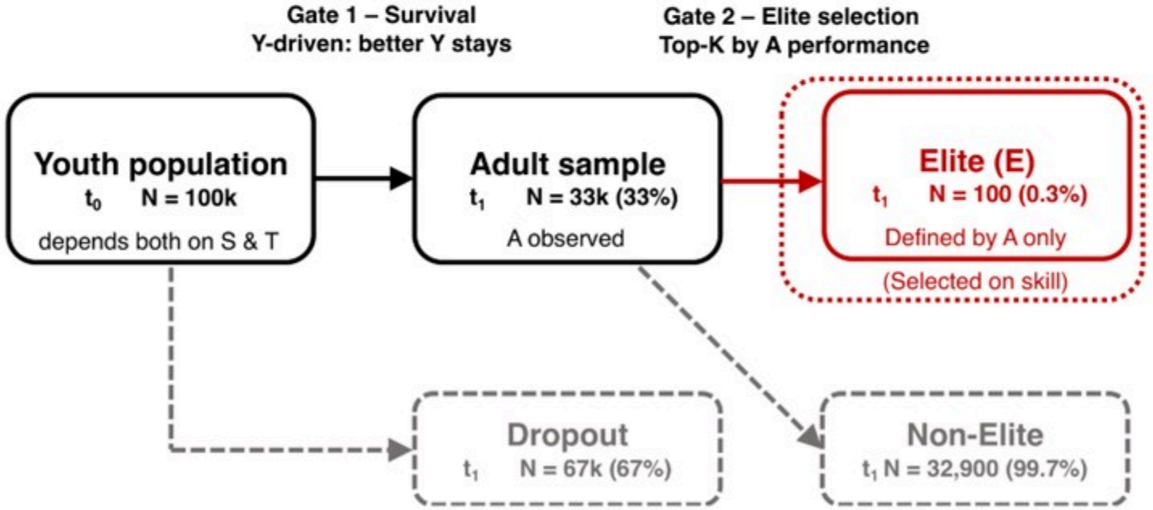
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B Selection gates (youth → adult → elite)



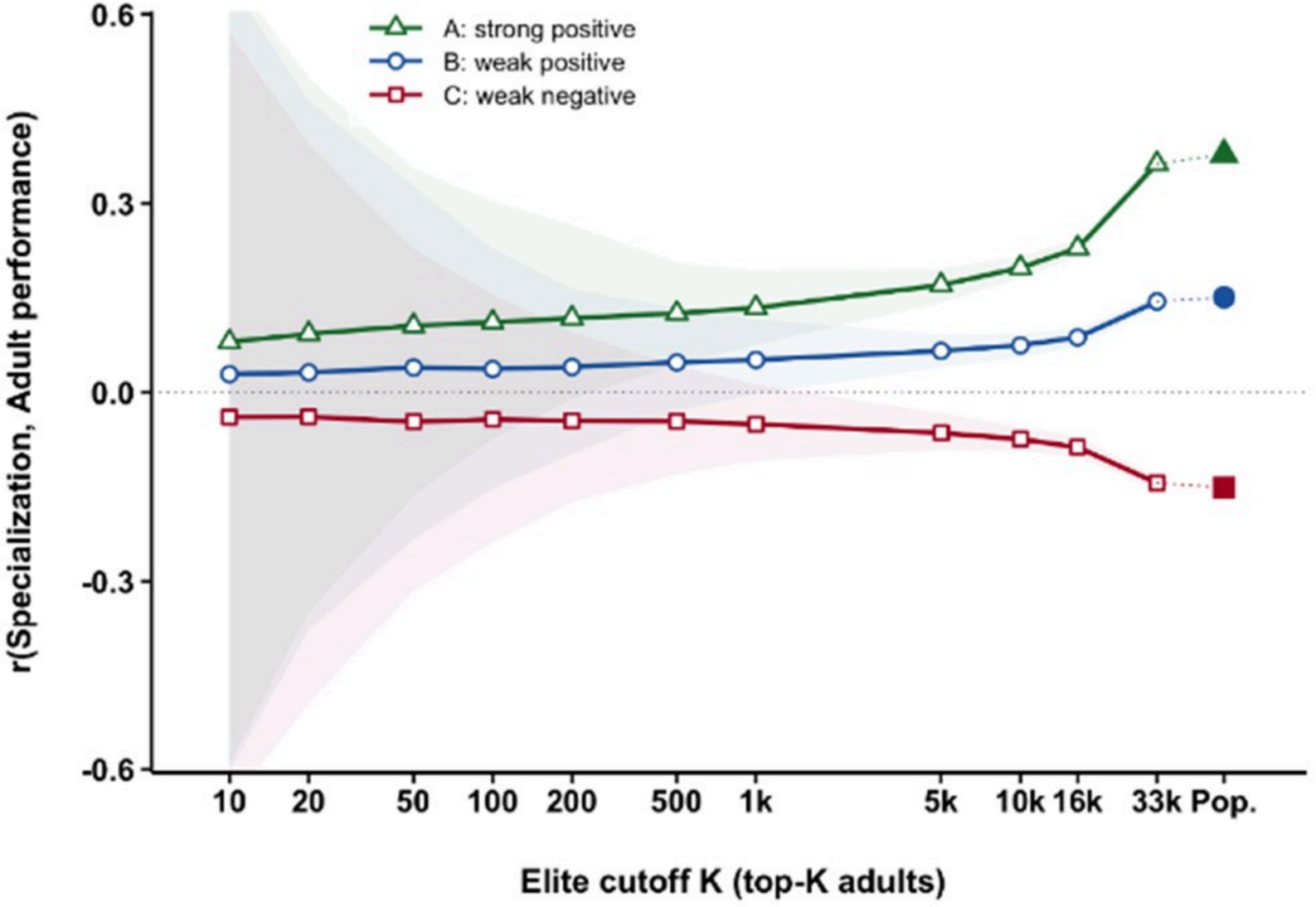
Simulation: Strength of the youth to adulthood effect

Scenario A: strong positive >0.3
Scenario B: weak positive = 0.2
Scenario C: weak negative = -0.2

In the full survivor sample the slopes clearly separate the effects.

At top-50 to top-100 (typical world-class study sizes), **35–45% of simulated cohorts recover the wrong sign**

Reliable separation needs an elite sample of ~**5,000**

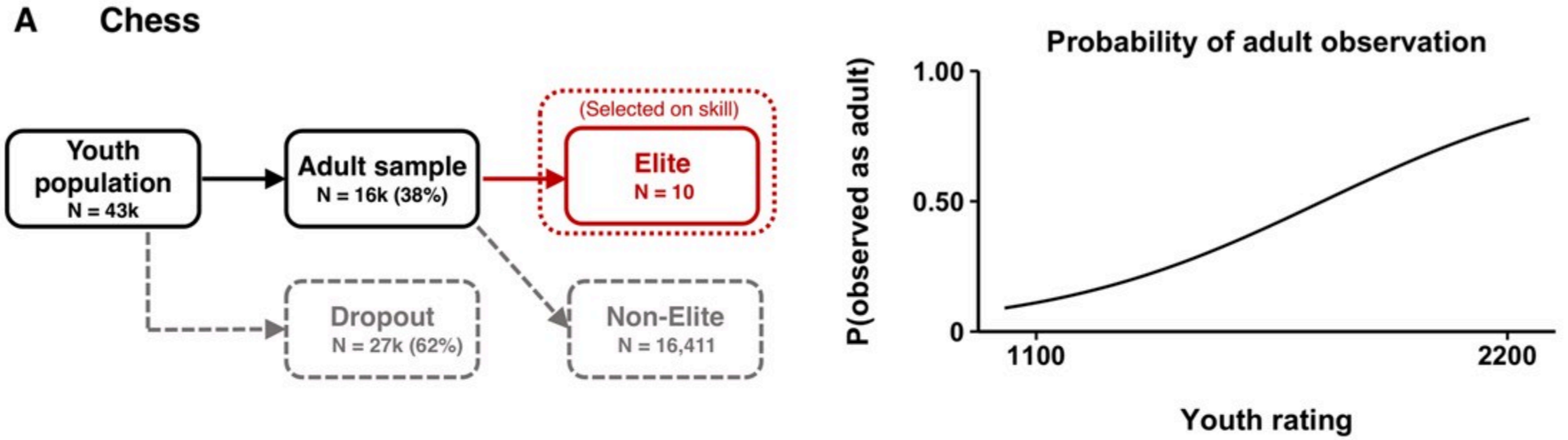


Empirical data: Gate 1 (who survives into adulthood)?

Longitudinal records — chess ratings and women's 1500 m

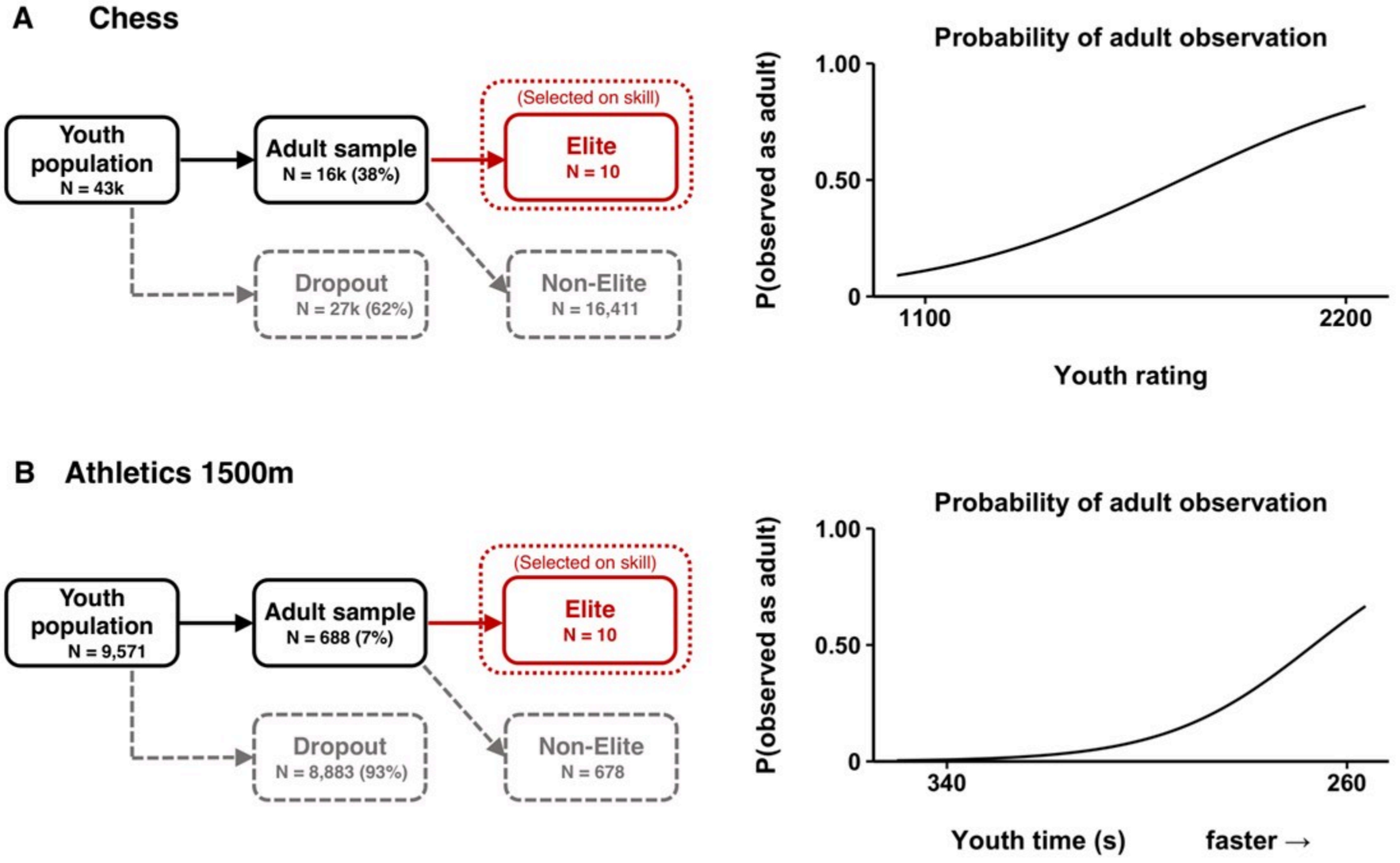
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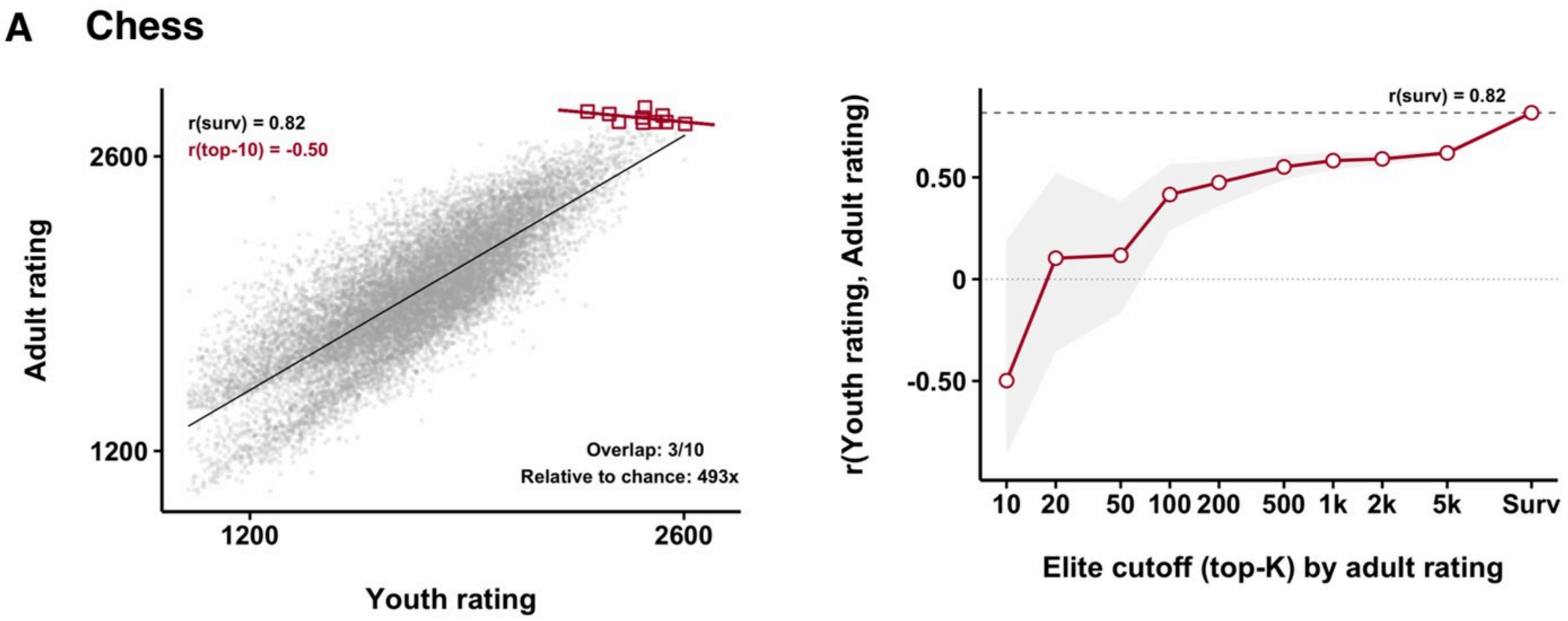
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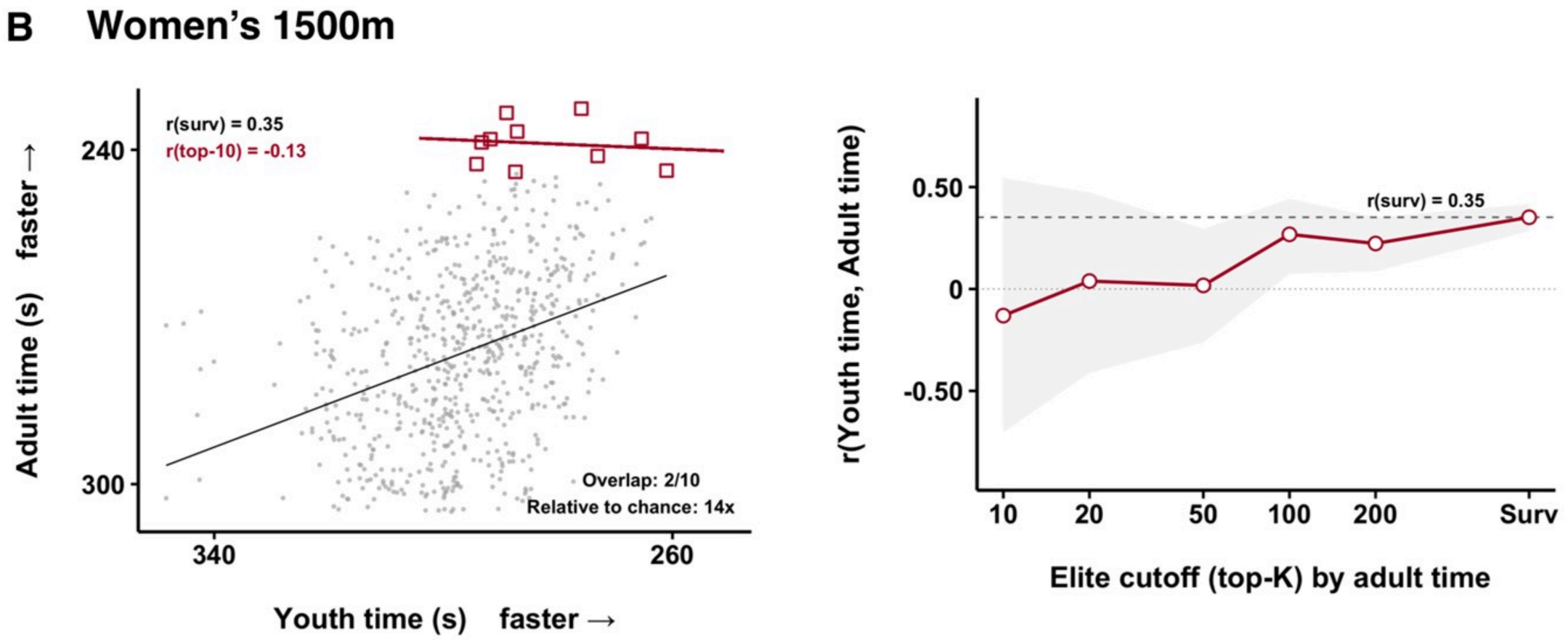
Empirical data: Gate 2 (what happens to the top-K people)?

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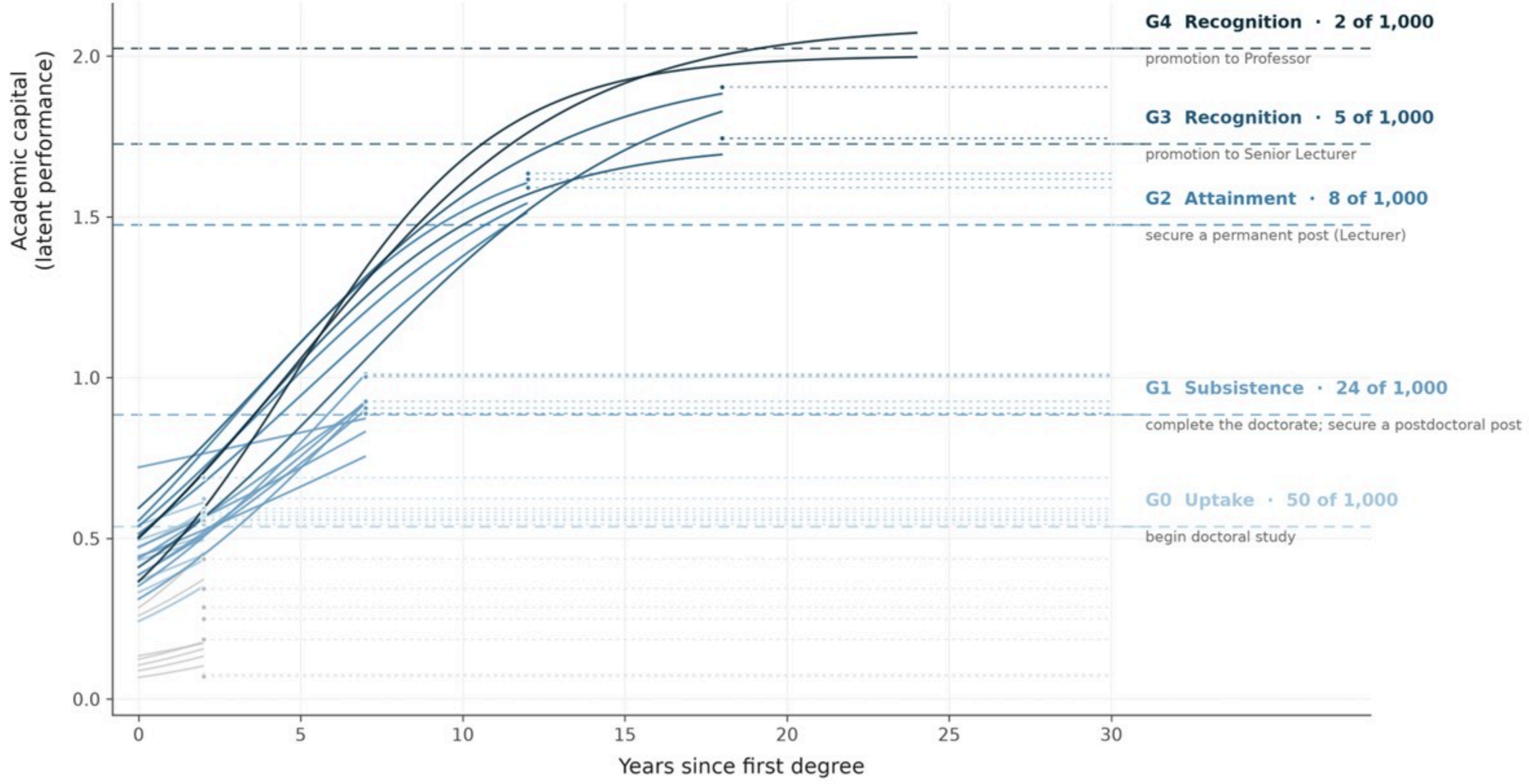
Longitudinal records — chess ratings and women's 1500 m



What to do about it: don’t ask professors how they became professors

What to do about it: Expertise development as sequential gate passage

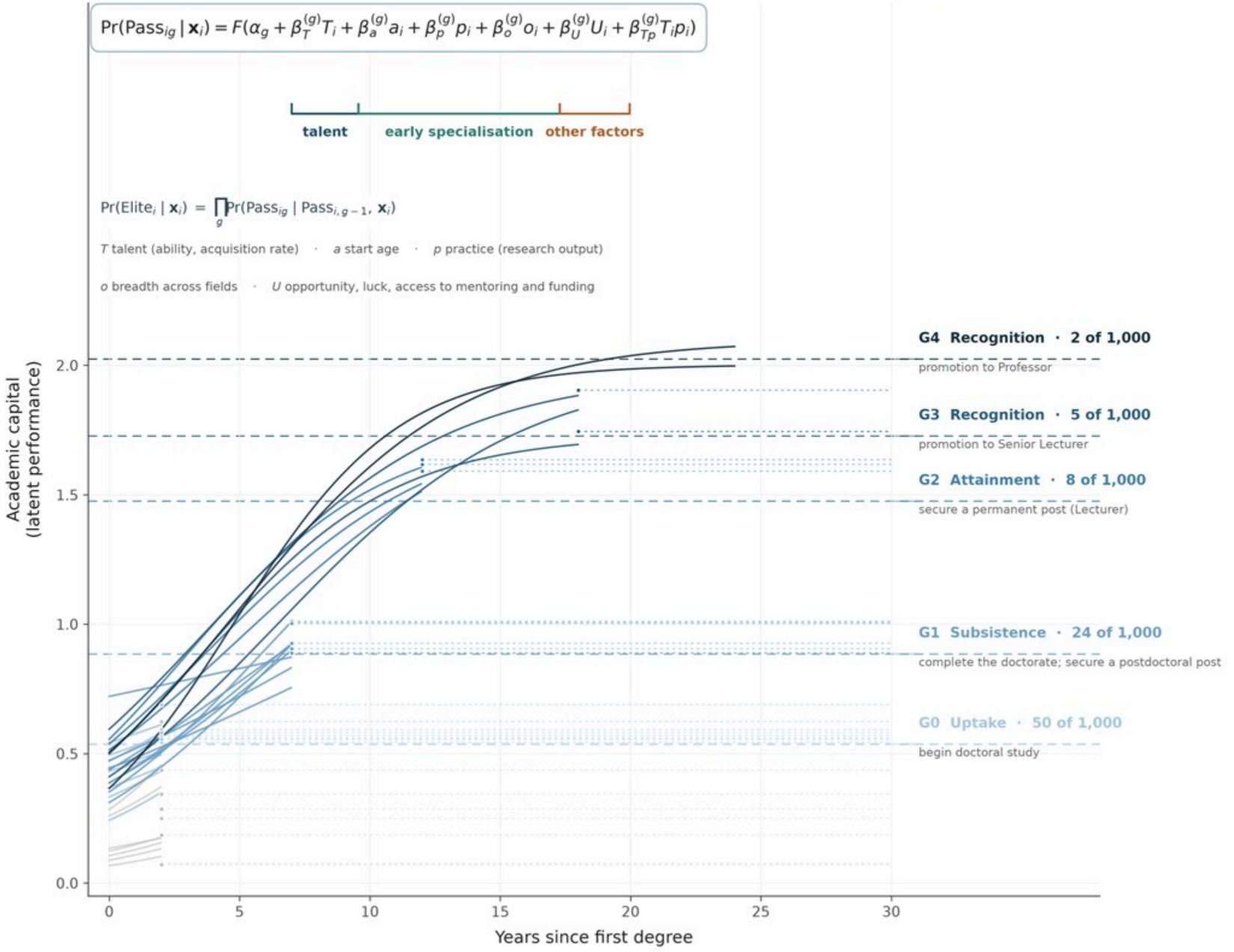
Who becomes an expert = who keeps passing gates; different factors matter at different gates



What to do about it: Expertise development as sequential gate passage

Who becomes an expert = who keeps passing gates; different factors matter at different gates

1. Identify when and where the gates select people
2. Model the interplay of factors that change the likelihood of gate passing
3. Feed-forward outcomes to correct assumptions at the next gate

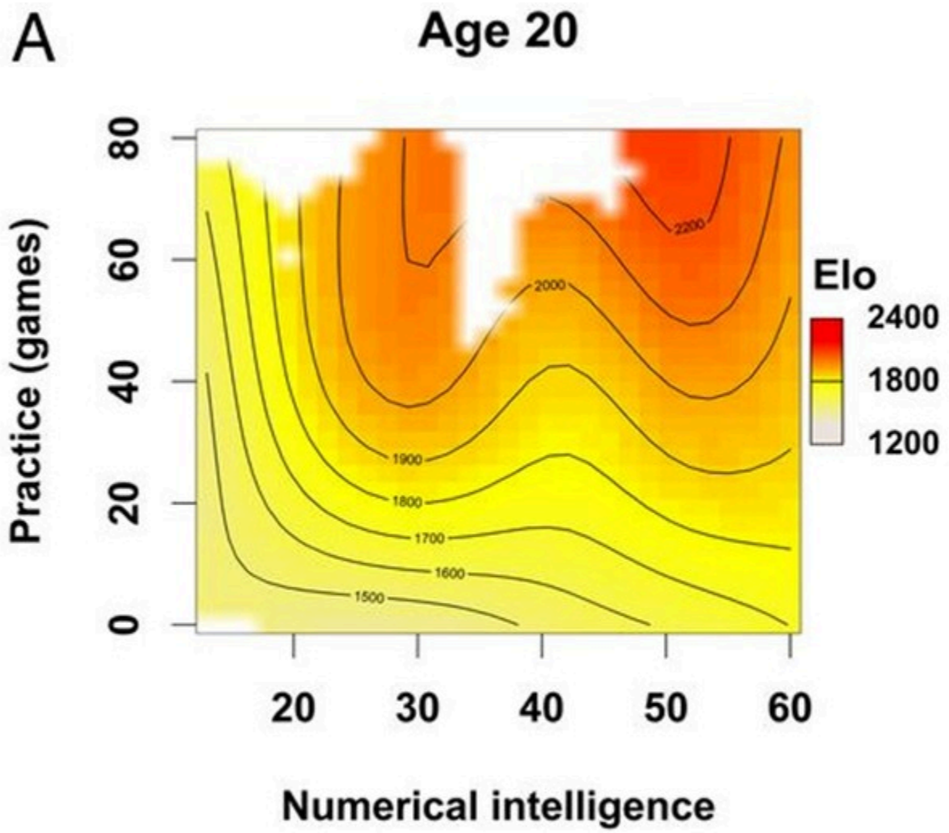


Looking back at my work: (Vaci et al, 2019)

More numerically able chess players are more likely to pass at least some of the gates and continue participating and improving

The joint influence of intelligence and practice on skill development throughout the life span. Proceedings of the National Academy of Sciences

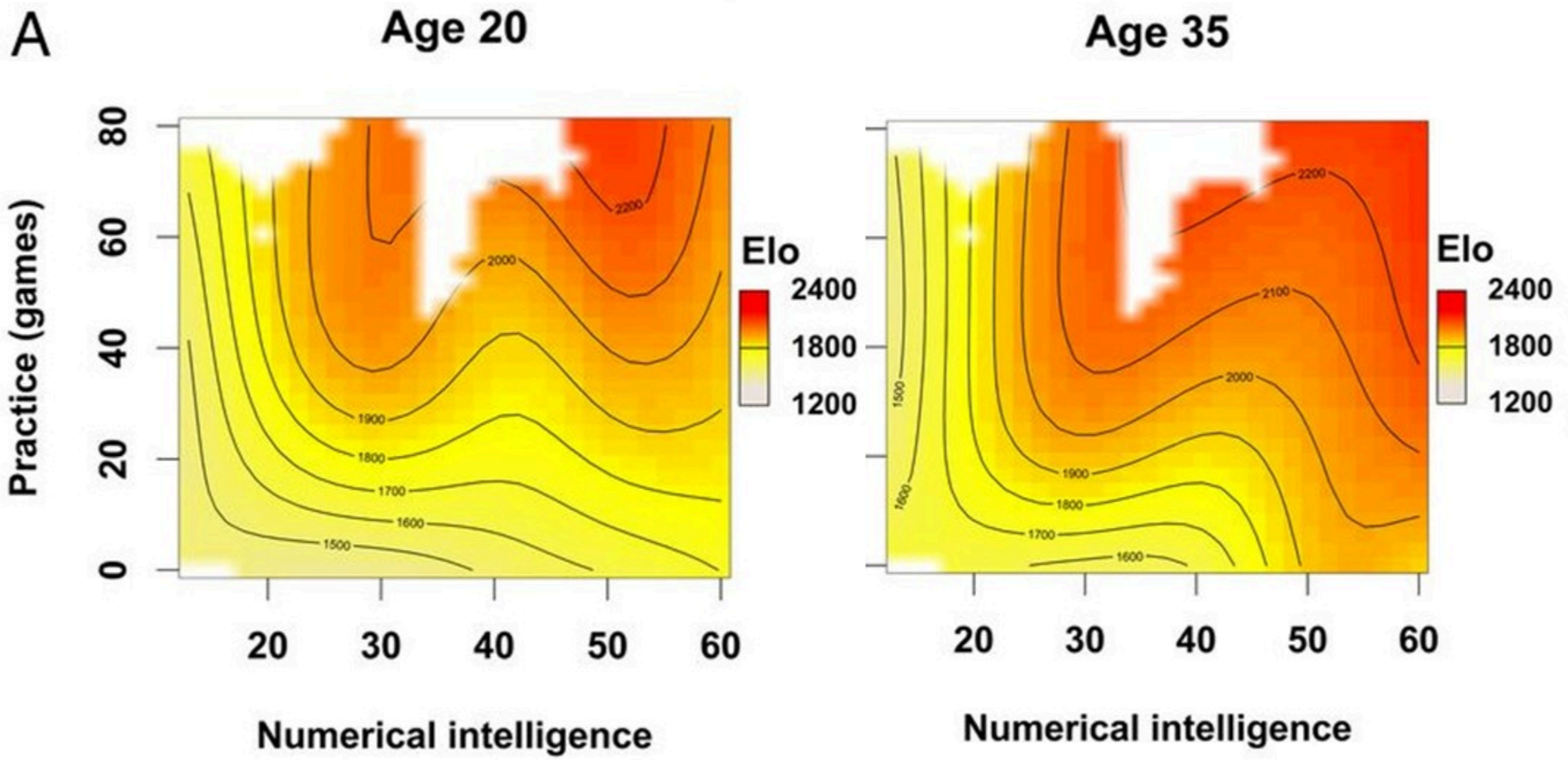
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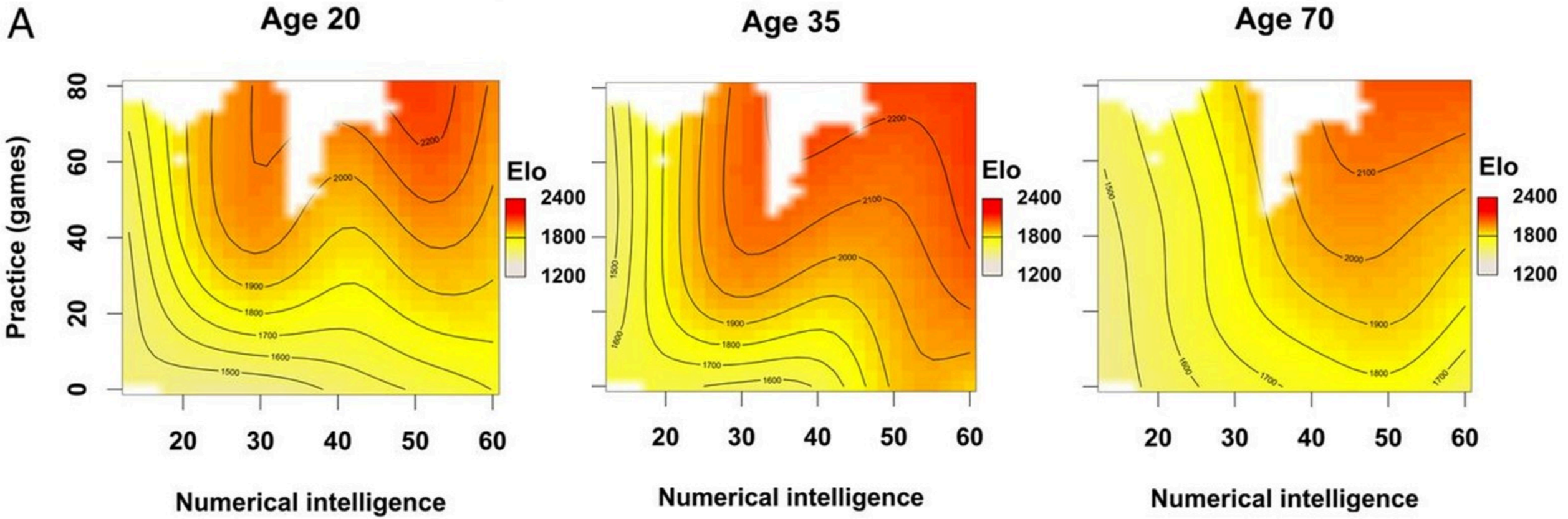
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Practical aspects: studying elites without being misled

Four reporting defaults + a selection-aware toolbox

1. Decide what you want to learn

Prediction within a pipeline ≠ population-level developmental inference — design for the goal you have

2. Define the cohort, map the gates

Start from an entry cohort; report who was lost at each gate (problem: most selection happens before the data exist)

3. Show threshold dependence

Report K-graphs, not one cutoff; test overlap against chance, not raw percentages

4. Match the claim to the sample

'Among those still competing at 22…' — selected-sample findings describe the selected group

Toolbox

Inverse-probability weighting (Hernán & Robins, 2020)

Sample-selection models (Heckman, 1979)

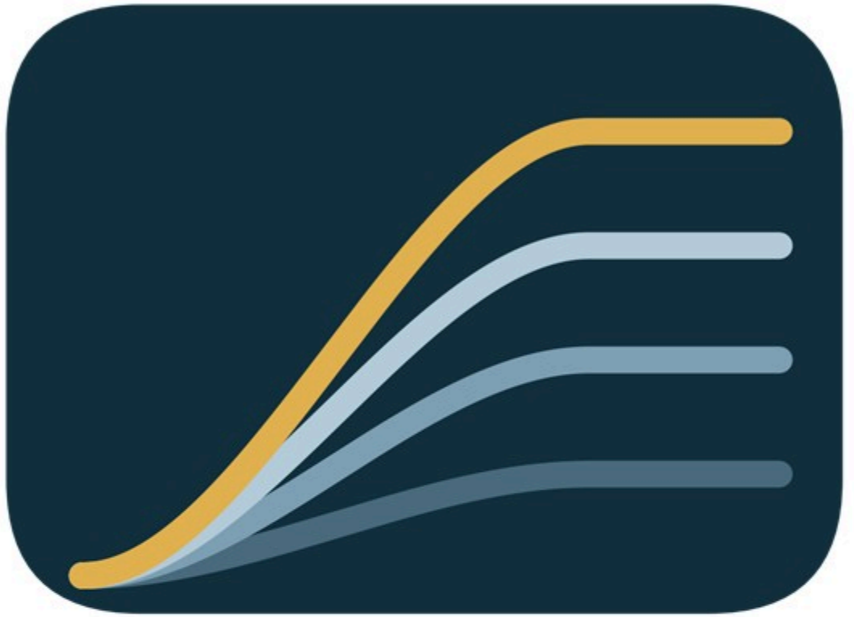
Range-restriction corrections (Sackett & Yang, 2000)

Negative controls (Lipsitch et al., 2010)

Admission lotteries among borderline candidates

Pathway tracking from a defined baseline

Questions?



Thank you

the gates we pass, and the ones we don’t